EE2FH4
Electromagnetics I

Term II, January – April 2026

MATLAB Examples and Exercises (Set 12)

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Example: A parallel-plate is filled with a nonuniform dielectric characterized by $\varepsilon_r = 2 + 2\times10^6 x^2$, where x is the distance from the lower plate in meters. If $S = 0.02$ m^2 and $d = 1.0$ mm, find the capacitance C. Write a MATLAB program that finds the energy stored in this capacitor if the charge on the positive plate is $Q = 4.0\times10^{-9}$ C. Use the formula $W_E = Q^2/2C$ to evaluate the capacitance and compare your results.

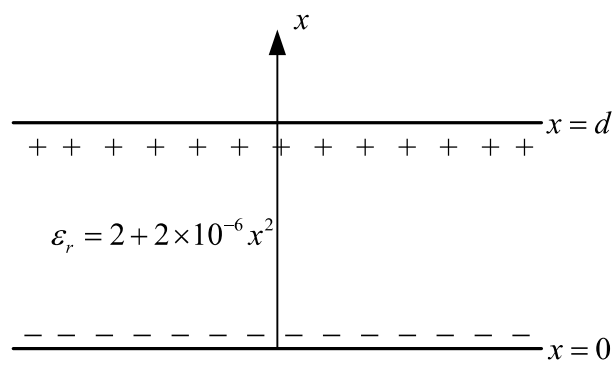

Figure 12.1 The geometry of the example of Set 12.

Analytical solution:
We can use the Q-method to find the capacitance. This can be done by first assuming a total charge of Q on the positive plate and then finding the potential difference V between the two plates. Finally, we can evaluate the capacitance by using $C=Q/V$. A total charge of Q is on the positive plate, and since the plate can be seen as an infinite plate ($\sqrt{S} \gg d$), we can simply assume a uniform charge density on the plates. The electric flux density is given by

$$\mathbf{D} = -\frac{Q}{S}\mathbf{a}_x$$

and the electric field intensity is given by

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_r\varepsilon_0} = -\frac{Q}{\varepsilon_r\varepsilon_0 S}\mathbf{a}_x .$$

By knowing the electric field intensity we can find the voltage difference between the two plates,

$$V = -\int_{x=0}^{x=d}\mathbf{E}\cdot d\mathbf{L} = -\int_{x=0}^{x=d} -\frac{Q}{\varepsilon_r\varepsilon_0 S}\mathbf{a}_x\cdot\left(\mathbf{a}_x dx\right) = \frac{Q}{\varepsilon_0 S}\int_{x=0}^{x=d}\frac{1}{2+2\times10^6 x^2}dx = \frac{Q}{\varepsilon_0 S}\times\frac{1}{\sqrt{10^{-6}}}\arctan\left(\frac{x}{\sqrt{10^{-6}}}\right)\Bigg|_{x=0}^{x=d}$$

$$= \frac{Q}{2000\varepsilon_0 S}\arctan\left(1000d\right)$$

Now, we can find the capacitance

$$C = \frac{Q}{V} = \frac{Q}{\dfrac{Q}{2000\varepsilon_0 S}\arctan\left(1000d\right)} = \frac{2000\varepsilon_0 S}{\arctan\left(1000d\right)} = \frac{2000\times\dfrac{1}{36\pi}\times10^{-9}\times0.02}{\arctan(1000\times10^{-3})} = 4.503\times10^{-10}\ \text{C}$$

MATLAB solution:

We will write a MATLAB program to find the energy stored in the capacitance then use the formula $W_E = Q^2/2C$ to evaluate the capacitance. The energy stored in the capacitor is given by

$$W_E = \frac{1}{2}\int_{\text{vol}} \varepsilon_r\varepsilon_0 E^2 dv = \frac{1}{2}\int_{\text{vol}} \frac{D^2}{\varepsilon_r\varepsilon_0} dv = \frac{1}{2}\int_{\text{vol}} \frac{D^2}{\varepsilon_r\varepsilon_0} dv$$

Consider a very thin layer of this capacitor. Since the relative dielectric ε_r varies only in the x direction, we can assume the dielectric is the same everywhere in the very thin layer. Also we note that the electric flux density is constant along the x direction. Therefore we can write a program that divides the capacitor into many thin layers and evaluate the energy stored in each layer. We then add all the energy stored in these layers together to obtain the total energy stored in the capacitor. By knowing the energy stored in the capacitor, we can calculate the capacitance by using $C = Q^2/2W_E$

MATLAB code:

```
clc; %clear the command line
clear; %remove all previous variables
 % initialize variables
eo=1e-9/(36*pi); % the permittivity in free space
Q=4e-9;          % charges on the positive plate
S=0.02;          % area of the capacitor
d=1e-3;          % thickness of the capacitor
Ds=Q/S;          % electric flux density
Number_of_x_steps=100;  %number of steps in the x direction
dx=d/Number_of_x_steps; %x increment
% perform calculations
W=0;  % initialize the total energy
for k=1:Number_of_x_steps
    x=0.5*dx+(k-1)*dx;  %current radius
    er=2+2*x*x*1e6;     %current relative permittivity
    dW=0.5*Ds*Ds*S*dx/(er*eo);  % energy stored in a thin layer
    W=W+dW;                     % get contribution to the total energy
end
C=Q^2/(2*W)
```

Running result:

```
C =

   4.5032e-010
```

Comparing the answers we see that our MATLAB solution and analytical solution are consistent.

Exercise: A very long coaxial capacitor has an inner radius of $\rho_{inner} = 1.0 \times 10^{-3}$ m and an outer radius of $\rho_{inner} = 5.0 \times 10^{-3}$ m. It is filled with a nonuniform dielectric characterized by $\varepsilon_r = 10^3 \rho$. Find the capacitance of a 0.01 m long capacitor of this kind. Write a MATLAB program that finds the energy stored in this capacitor if the charge on the inner plate is $Q = 5.0 \times 10^{-9}$ C . Use the formula $W_E = Q^2 / 2C$ to evaluate the capacitance again and compare your results.

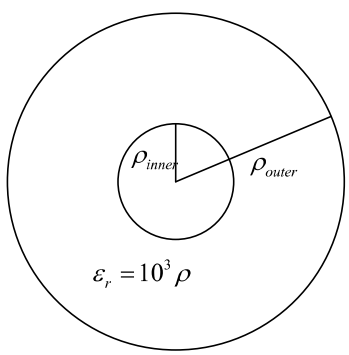

Figure 12.2 The cross section of the coaxial capacitor of the exercise of Set 12.